ACSC383 - System Requirements Specification

System Requirements Specification

1. System Architecture Requirements

1. The system shall be implemented as a **web-based application** accessible through modern web browsers.
2. The system shall follow a **client–server architecture**.
3. The system shall support a **mobile-first responsive design**.
4. The system shall separate concerns between:
   - Presentation layer (frontend)
   - Application logic layer (backend)
   - Data persistence layer (database)

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2. Authentication and Authorization Requirements

5. The system shall restrict access to authenticated users only.
6. The system shall require account registration using a **university-approved email domain**.
7. The system shall verify user accounts via email before activation.
8. The system shall securely authenticate users using encrypted credentials.
9. The system shall enforce session management and automatic session expiration.
10. The system shall authorize users such that:
    - Users may only modify their own data
    - Users may only manage their own items and transactions

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3. User Data Management Requirements

11. The system shall store user account information securely.
12. The system shall maintain a unique identifier for each user.
13. The system shall associate items and transactions with user accounts.
14. The system shall allow retrieval of user transaction history.
15. The system shall compute and store cumulative environmental impact per user.

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4. Item Management Requirements

16. The system shall allow authenticated users to create item listings.
17. The system shall require the following attributes for each item:
    - Title
    - Description
    - Category
    - Condition
    - Transaction type
18. The system shall support optional pricing for sale items.
19. The system shall support uploading and storing item images.
20. The system shall allow users to update or delete their own item listings.
21. The system shall maintain an item availability state:
    - Available
    - Reserved
    - Completed

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5. Item Discovery and Retrieval Requirements

22. The system shall retrieve and display only **available items** for browsing.
23. The system shall support keyword-based search.
24. The system shall support filtering by:
    - Category
    - Condition
    - Transaction type
25. The system shall prevent access to completed or reserved items for new transactions.

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6. Transaction Management Requirements

6.1 General Transaction Handling

26. The system shall create a transaction record when a transaction is proposed.
27. The system shall uniquely identify each transaction.
28. The system shall associate transactions with one or more items.
29. The system shall track the status of each transaction:
    - Pending
    - Accepted
    - Declined
    - Cancelled
    - Completed

6.2 Swap Transactions

30. The system shall allow swap transactions involving **two distinct items**.
31. The system shall validate that both items are:
    - Available
    - Owned by different users
32. The system shall reserve both items when a swap is accepted.
33. The system shall prevent either item from participating in other transactions while reserved.
34. The system shall require confirmation from both users to complete a swap.

6.3 Sale and Donation Transactions

35. The system shall allow sale and donation transactions involving **one item**.
36. The system shall reserve the item once the owner accepts the request.
37. The system shall prevent multiple active transactions for the same item.

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7. Transaction Completion Requirements

38. The system shall require user confirmation before marking a transaction as completed.
39. The system shall delist items upon successful transaction completion.
40. The system shall archive completed transactions for historical reference.

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8. Environmental Impact Calculation Requirements

42. The system shall estimate environmental impact for each completed transaction.
43. The system shall calculate waste diverted based on item category averages.
44. The system shall calculate CO_2 emissions saved based on avoided production estimates.
45. The system shall store environmental impact values per transaction.
46. The system shall aggregate environmental impact values per user.
47. The system shall display environmental impact summaries to users.

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9. Notification System Requirements

48. The system shall generate notifications for key transaction events.
49. The system shall notify users when:
    - A transaction is proposed
    - A transaction is accepted or declined
    - A transaction status changes
50. The system shall store notifications for later viewing.
51. The system shall mark notifications as read or unread.

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10. Data Integrity and Consistency Requirements

52. The system shall ensure item and transaction states remain consistent.
53. The system shall prevent race conditions when multiple users act on the same item.
54. The system shall enforce atomic updates for transaction acceptance and reservation.
55. The system shall roll back incomplete operations in case of failure.

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11. Security Requirements

56. The system shall hash and salt user passwords.
57. The system shall validate and sanitize all user inputs.
58. The system shall prevent unauthorized data access.
59. The system shall restrict file uploads to approved image formats.
60. The system shall protect against common web vulnerabilities.

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12. Performance and Reliability Requirements

61. The system shall respond to user requests within acceptable time limits.
62. The system shall support multiple concurrent users.
63. The system shall ensure availability during normal operating conditions.
64. The system shall recover gracefully from unexpected errors.

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13. Maintainability and Extensibility Requirements

65. The system shall use modular, maintainable code structures.
66. The system shall allow future extension of item categories.
67. The system shall allow future refinement of environmental impact calculations.

68. The system shall support database schema evolution without data loss.

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14. Constraints

69. The system shall be implemented using standard web technologies.
70. The system shall not integrate external payment services.
71. The system shall not provide real-time messaging between users.
72. The system shall operate exclusively within the university community.